Sapienza University of Rome
Faculty of Mathematical, Physical and Natural Sciences
Master's Degree in Mathematics

Master's Thesis Summary

Convexity properties in Symplectic Geometry: Moment Mapping and Toric Manifolds

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Introduction and Geometric Context

This thesis explores symplectic geometry, providing a rigorous analysis of Hamiltonian actions of Lie groups on symplectic manifolds, focusing on the topological and geometric properties of the moment map. The theoretical framework treated here draws its original motivation from classical mechanics and Noether's Theorem, which establishes that every continuous symmetry of a physical system corresponds to a conserved quantity. In the language of symplectic topology, these symmetries are formalized through the action of a Lie group, and the conserved quantities (such as linear or angular momentum) are unified into a single, powerful geometric object: the moment map.

Chapter 1 is dedicated to establishing the necessary theoretical foundations. It provides an overview of symplectic geometry, starting from non-degenerate skew-symmetric bilinear forms on vector spaces, moving to the definition of a symplectic manifold and Hamiltonian vector fields. The chapter also recalls fundamental results like Darboux's Theorem, which guarantees the absence of local symplectic invariants, and introduces basic concepts of Lie groups and Lie algebras theory, which are essential for treating continuous symmetries.

The Moment Map and Local Analysis

Chapter 2 delves into the core of the theory by formally introducing the Hamiltonian action and the moment map $\Phi : X \to \mathfrak{g}^*$, where X is the symplectic manifold and $\mathfrak{g}^*$ is the dual of the Lie algebra of the group G acting on it. The fundamental property of equivariance of the moment map with respect to the group action on the manifold and the coadjoint action on the dual of the Lie algebra is analyzed. The vital connection with mechanics is expressed by Hamilton's equation: the component of the moment map evaluated along a Lie algebra direction acts exactly as the Hamiltonian function for the fundamental vector field generated by that same direction.

The second part of the chapter focuses on the local behavior of the moment map around the fixed points of the group action. By exploiting a suitable invariant Riemannian metric and the exponential map, the system is linearized. Through an equivariant version of Moser's trick, a local convexity theorem is proven: the image of a neighborhood of a fixed point under the moment map takes the form of a convex polyhedral cone generated by the weights of the isotropy representation of the linearized action.

Global Convexity: The Atiyah-Guillemin-Sternberg Theorem

In Chapter 3, the focus shifts from local to global analysis, culminating in the proof of the celebrated Atiyah-Guillemin-Sternberg Convexity Theorem. This theorem states that, given a Hamiltonian action of a torus (a compact and abelian Lie group) on a compact and connected symplectic manifold, the global image of the moment map is a convex polytope. Furthermore, the vertices of this polytope correspond exactly to the images of the fixed points of the torus action.

To achieve this result, the thesis introduces a detour into Morse-Bott theory. It is shown that the components of the moment map are Morse-Bott functions whose critical manifolds possess strictly even indices. This topological rigidity guarantees the absence of local minima or saddles of odd index, forcing the function to have a unique local maximum which necessarily coincides with the global maximum.

The chapter concludes by addressing the generalization of the theorem to actions of non-abelian Lie groups. In this context, the global image of the moment map loses its convexity due to the structure of the coadjoint orbits. However, as demonstrated by Kirwan's Theorem, intersecting the moment map's image with a positive Weyl chamber miraculously recovers convexity, yielding once again a convex polytope. This result highlights how profoundly the group symmetries govern the geometry of the entire manifold.

Classification of Toric Manifolds (Delzant's Theorem)

Chapter 4 represents the culmination of the work, focusing on the classification of symplectic toric manifolds. Such manifolds constitute the case of maximal possible symmetry, where the dimension of the acting torus is exactly half the dimension of the symplectic manifold (i.e., it acts with orbits of maximal dimension for a Hamiltonian action).

For these manifolds, the convex polytope generated by the moment map possesses extraordinary geometric rigidity properties and is known as a Delzant polytope (characterized by being simple, rational, and smooth). The thesis explores symplectic reduction (Marsden-Weinstein Theorem) and the symplectic cutting method introduced by Lerman, powerful tools that allow the construction of new symplectic manifolds by taking quotients.

The work closes with a complete and detailed proof of Delzant's Classification Theorem. This milestone theorem establishes a perfect one-to-one correspondence between symplectic toric manifolds (up to equivariant symplectomorphisms) and Delzant polytopes (up to translations). In other words, it demonstrates that the entire topological and symplectic structure of the manifold is rigidly and uniquely encoded in the simple combinatorics of its associated polytope. This result seals the elegance of the moment map as the ultimate bridge between differential geometry, Lie theory, and combinatorial algebra.

References

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